

AHMED GOMAA EL-SAYED

Data Analyst

Egypt | +20 1019672878 | ahmedgomaelsayed@gmail.com | [LinkedIn Profile](#) | [GitHub Profile](#) | [Portfolio Link](#)

PROFESSIONAL SUMMARY

Detail-oriented Data Analyst with hands-on portfolio experience turning complex datasets into actionable business insights. Proficient in Excel, SQL, Python, and Power BI, with a strong focus on Customer Behavior Analysis, HR Workforce Analytics, Retail Omnichannel Audit, and Business Intelligence. Skilled in Data Cleaning, Data Validation, Data Storytelling, Anomaly Detection, ETL Pipeline Architecture, KPI Reporting, and Analytical Thinking to drive executive decision-making. Core competencies in data accuracy, recordkeeping, and reconciliation are directly transferable to inventory and stock reconciliation, sales and business reporting, and general office/administrative operations. Academic background in psychology provides a unique edge in interpreting behavioral and operational patterns, customer friction, and communicating clear, impactful findings to stakeholders. Completed military service and immediately available for full-time roles.

CORE COMPETENCIES

Data Analysis & ETL: Data Accuracy & Validation, ETL Pipeline Architecture, Anomaly Detection & Anomaly Rules, Data Cleaning, Identity Resolution

Reporting & Modeling: Business Intelligence, KPI Tracking, Star Schema Modeling, Dashboard Design, Business & Sales Reporting

Business Acumen: Customer Behavior & Behavioral Analytics, Margin Leakage Detection, Inventory & Stock Reconciliation, Record-Keeping, Stakeholder Communication

TECHNICAL SKILLS

- **Data Analysis:** Microsoft Excel, Power Query, Pivot Tables, Power Pivot, Dashboard Design, Data Cleaning, Data Validation, Data Entry & Documentation
- **SQL:** PostgreSQL, MySQL, CTEs, Window Functions (ROW_NUMBER, LAG, LEAD), Joins, CASE WHEN, Aggregations, ETL Staging
- **Python:** Pandas, NumPy, Matplotlib, Seaborn, Isolation Forest, IQR Anomaly Detection, Exploratory Data Analysis (EDA)
- **Business Intelligence:** Excel Dashboards, KPI Reporting, Business Reporting, Power BI, DAX, Star Schema Data Modeling, Data Quality SLAs

PROJECTS

Enterprise Omnichannel Audit & Margin Leakage Detection

Tools: SQL (PostgreSQL, CTEs, Window Functions), Python (Pandas, Isolation Forest, IQR), Power BI, DAX, Star Schema | **Link:** github.com/Ahmed1870/enterprise-omnichannel-audit

- Executed a simulated enterprise retail audit across Store, Web, Mobile App, and Marketplace channels on 108,000 transactions (\$6.98M Gross Sales) to identify margin erosion and revenue leakage.
- Engineered an end-to-end data pipeline and ETL workflow: automated CSV profiling, SQL staging, ISO/Unix timestamp normalization, customer identity resolution (email/phone typography), category mapping, and multi-currency (FX) conversion.
- Built a Star Schema dimensional model (FactTransactions linked to 6 Dimension tables: DimDate, DimCustomer, DimProduct, DimChannel, DimPromotion, DimCurrency) to power Power BI semantic modeling and executive reporting.
- Applied advanced SQL data engineering (CTEs, ROW_NUMBER, LAG, LEAD) to analyze cohort retention, return-risk segmentation, and policy compliance (>180-day window).
- Performed Python-based anomaly detection (IQR & Isolation Forest) to pinpoint promotional abuse (\$132.6K leakage proxy, 1.5% abuse rate) and suspicious returns (\$125.7K proxy, 44% risk rate).
- Leveraged psychological behavioral insights (wardrobing behavior, cognitive friction) to design selective friction rules, progressive verification, and auditable controls that protect revenue without harming customer retention.
- Developed interactive Power BI dashboards (Executive, Fraud Risk, Returns) with custom DAX measures tracking Net Sales (\$6.27M), Realized Margin (\$2.46M), and Return Friction Index to drive business governance.

Customer Behavior & RFM Analysis

Tools: SQL, Python (Pandas, NumPy, Matplotlib), Excel, Power Query, Power BI | **Link:** github.com/Ahmed1870/Customer-Behavior-RFM-Analysis

- Completed an end-to-end data analysis project based on the Online Retail Dataset (Kaggle/UCI), executing comprehensive data cleaning, data validation, and customer-level aggregation to deliver actionable business intelligence.
- Cleaned and processed more than 540,000 retail transactions to ensure data integrity and reliability.
- Calculated Recency, Frequency, and Monetary (RFM) metrics and built customer-level analytical tables using advanced SQL queries (including Joins, CTEs, CASE WHEN logic, Window Functions, and Aggregations).
- Segmented customers into actionable cohorts (Champions, Loyal Customers, Potential Loyalists, At Risk, About To Sleep, and Lost Customers) to design highly-targeted marketing efforts.
- Translated customer segments into tailored retention strategies, identifying a potential recovery of up to 15% in at-risk revenue by targeting dormant cohorts.
- Performed exploratory data analysis (EDA) using Python (Pandas, NumPy, Matplotlib) to uncover underlying purchasing patterns.
- Built an interactive Excel dashboard to monitor KPIs and track core business metrics dynamically.
- Generated business recommendations for increasing customer retention and revenue, documenting the complete workflow on GitHub.

Employee Attrition Analysis (HR Analytics)

Tools: SQL, Python (Pandas, NumPy), Power BI, DAX | **Link:** github.com/Ahmed1870/HR-Attrition-Data-Analysis

- Analyzed a 5,000-employee HR dataset with a 15% attrition rate to uncover the workforce risk factors driving employee turnover.
- Cleaned and validated raw HR records in SQL and engineered features in Python (income quartiles, commute-distance bands, overtime flag) ahead of modeling.
- Built an interactive Power BI dashboard with DAX measures breaking attrition down by department, overtime status, income quartile, commute distance, and job role.
- Identified overtime as the strongest attrition driver (29.3% vs. 8.7%, a 3.4x gap) and flagged Executive and Analyst roles as unexpected high-risk segments.
- Delivered targeted retention recommendations — overtime caps in high-risk departments, compensation review for the lowest income quartile, and a dedicated retention track for high-risk roles.

EDUCATION

Bachelor of Arts – Psychology — Graduated: 2024

Faculty of Arts, Fayoum University, Egypt

Relevant Coursework: Research Methods, Statistical Analysis, Behavioral Data Interpretation, Academic Report Writing

CERTIFICATIONS & TRAINING

- Google Data Analytics Professional Certificate — Google via Coursera (2024)
- Excel Skills for Business — Macquarie University via Coursera (2024)
- SQL for Data Science — UC Davis via Coursera (2024)
- Data Analysis with Python — IBM via Coursera (2024)
- Microsoft Power BI Desktop for Business Intelligence — Udemy (2024)

ADDITIONAL INFORMATION

- **Languages:** Arabic (Native), English (Professional Working Proficiency)
- **Military Service:** Completed – No obligations